THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

1. Executive summary

Floods displace millions of people in India each year. The limiting factor in a rescue is locating survivors, not reaching them.

Three one-metre, 6.4 kg multirotor aircraft search flooded ground as a coordinated group under one operator, identify people from the air, report their position to a metre, and deliver a relief kit to each. No mobile network or internet is required.

Sizing, error budgets and flight software are complete; coordination is demonstrated across three simulated autopilots. No aircraft has flown, so the figures here are calculated.

2. The problem

Boat teams search slowly, at risk, and cannot see rooftops. Helicopters are scarce and committed to evacuation. Conventional drones need one pilot each, depend on infrastructure the flood has removed, and cannot assist a survivor once found.

Detection is the harder problem. A person in floodwater presents head and shoulders, 40 cm across. Detection models take a fixed, small input, so the usual approach shrinks the image and the target with it, to a size at which published accuracy is 29–61 %. A smaller model is worse: smaller models take smaller inputs.

No dataset annotates people in flood imagery. Thermal imaging fails in water, where skin equilibrates toward water temperature.

3. Our solution

System. The operator draws a search region of any shape. The software splits it into three balanced sub-regions and plans a path over each, verified on thirteen non-convex shapes with no path leaving the region and imbalance below 0.02 %. The aircraft fly autonomously; the test harness shows structurally that no command is sent after the mission begins. Detection and geolocation run on board. Each aircraft carries four kits, descends, nulls its groundspeed over a survivor and releases one. Link loss, low battery and operator abort converge on a single return-and-land behaviour.

Detection. The image is not shrunk. Each frame is divided into 48 overlapping tiles at native resolution, and the survey altitude is 40 m rather than 60 m. The target becomes **39 pixels** rather than 13, nine times the area, outside the band where detector performance collapses. Either change alone leaves it inside.

The cost is four times the computation: 48 tiles at 2 Hz is 96 inferences per second, against 130–160 measured for the accelerator. Mission duration is shorter than at 60 m, the lower climb outweighing the extra transects.

Open question. Confirming a survivor across twelve frames needs 3 Hz, which exceeds that capacity. A two-stage filter, discarding the 87 % of tiles that are open water, would recover it. A filter that discards a tile containing a person fails without indication, so it will be evaluated against public data before adoption.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that displays mission state and is architecturally incapable of commanding the aircraft.
3. Autonomy software — partitioning, path planning, allocation, failsafes — with an automated test suite.
4. Measured detection accuracy and a verdict on the two-stage filter.
5. A publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
6. Source code and results, published in either outcome.

5. Benefits

- **Operational.** Ten hectares in under eight minutes, returning coordinates rather than a video feed needing continuous attention.
- **Economic.** The complete system costs less than one hour of helicopter search time, travels by road, and is operated by two people.
- **Institutional.** A thrust stand, certified scale, ground station, tested codebase and trained students remain with the department.
- **Strategic.** Indian suppliers wherever they meet the specification; domestic lead times are half those of imports.

6. Resources needed

Funding is requested in 32 small steps rather than as one grant. Each step buys one named set of parts and is released only once the previous step has produced a stated result. If a step fails, the programme stops there and nothing beyond it has been spent.

The order follows engineering dependency: one motor is measured before the other eleven are bought, and the first aircraft flies a complete mission before the second is begun.

Phase schedule

The third column is the amount released. The fourth is what must already be true before it is released.

#	Objective	INR	Released only after
1	Establish thrust and mass measurement capability	26,196	Approval
2	Verify static thrust against the 3.18 kgf requirement	5,175	Instruments commissioned
3	Aircraft 1 — autopilot and control link	25,990	**Thrust verified**
4	Aircraft 1 — onboard computer and AI accelerator	21,850	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning (RTK receiver)	27,892	Compute stack operating
6	Aircraft 1 — camera and radios	29,440	Position fix acquired
7	Aircraft 1 — airframe fabrication	20,144	Avionics integrated
8	Aircraft 1 — battery pack assembly	29,299	Airframe fabricated
9	Aircraft 1 — speed controllers, propellers, release servos	27,098	Pack bench-discharged
10	Aircraft 1 — propulsion completed	15,525	Drive train installed
11	Charging and manual override for flight test	25,300	Aircraft assembled and weighed
12	Field equipment: consumables, sun hood	17,274	Charging verified
13	Aircraft 2 — autopilot and control link	25,990	**Aircraft 1 flew a full mission**
14	Aircraft 2 — onboard computer and AI accelerator	21,850	Autopilot bench-tested
15	Aircraft 2 — centimetre positioning (RTK receiver)	27,892	Compute stack operating
16	Aircraft 2 — camera and radios	29,440	Position fix acquired

#	Objective	INR	Released only after
17	Aircraft 2 — airframe fabrication	20,144	Avionics integrated
18	Aircraft 2 — battery pack assembly	29,299	Airframe fabricated
19	Aircraft 2 — speed controllers, propellers, release servos	27,098	Pack bench-discharged
20	Aircraft 2 — propulsion completed	20,700	Drive train installed
21	Aircraft 3 — autopilot and control link	25,990	**Two aircraft, separation verified**
22	Aircraft 3 — onboard computer and AI accelerator	21,850	Autopilot bench-tested
23	Aircraft 3 — centimetre positioning (RTK receiver)	27,892	Compute stack operating
24	Aircraft 3 — camera and radios	29,440	Position fix acquired
25	Aircraft 3 — airframe fabrication	20,144	Avionics integrated
26	Aircraft 3 — battery pack assembly	29,299	Airframe fabricated
27	Aircraft 3 — speed controllers, propellers, release servos	27,098	Pack bench-discharged
28	Aircraft 3 — propulsion completed	20,700	Drive train installed
29	Ground base receiver, the source of the corrections	27,892	First flight completed
30	Stable, repeatable base mounting	4,600	Base receiver acquired
31	Deployable ground segment	28,750	Base station operating
32	Statutory registration	5,750	Ground segment complete
	Total	**7,43,004**	

Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Phase 13 begins the second aircraft only after the first has flown a complete mission, so the largest blocks of expenditure fund replication of a demonstrated configuration rather than a projection.

Component selection

The schedule gives the amount *released* per phase. The tables below give the parts each phase *buys*. The two differ by a fixed rule: a released amount is the parts cost plus 22 % customs duty on the imported share, plus 18 % GST, plus 15 % contingency. Across the programme that is INR 5,86,400 of parts against INR 7,43,004 released, a factor of 1.27. Every row below carries the phase number that buys it, so any line can be traced to its approval.

The seven lines immediately below are current retail listings from named Indian suppliers, so they are firm. Every other line in this section is a quotation or an estimate from a comparable part, and those are where the total is most likely to move.

Item	Price seen	Supplier, and what we budgeted
Propellers	3,250/pair	Robokits India; budgeted at this figure
Battery charger	5,836	Indian Hobby Center, ToolkitRC M6D
Safety-pilot radio	6,955	IndiaMART, RadioMaster Boxer ELRS
AI accelerator	9,950–11,309	multiple Indian retailers; budgeted 11,000
Motor	3,378–6,690	Robokits to IndiaMART; budgeted 4,500 mid-range
Battery cells	405 imported	Robokits, Molicel P45B. **Budgeted 700** for the domestic supplier, which keeps one of four Indian-sourced lines
Onboard computer	6,955–8,530	several Indian retailers; budgeted 8,000

The flight controller, GNSS receiver, camera, speed controllers and mesh radio are carried at quotation. The GNSS line is the one to confirm first: the supplier has confirmed RTK rover operation, which was the open question, but neither candidate receiver is publicly priced, so INR 18,000 is an estimate and phases 5, 15, 23 and 29 move with it.

Terms used below. *RTK*: satellite positioning corrected against a second receiver on a surveyed point, giving centimetre rather than metre accuracy. *6S3P*: eighteen cells, six in series for voltage and three in parallel for capacity. *kgf*: thrust written as the weight it would lift. *TOPS*: trillion operations per second, the accelerator's throughput. *ESC*: the electronic speed controller between battery and motor. *BMS*: the board that balances and protects the cells.

Measurement instruments — phases 1–2 INR 24,500 of parts

Ph.	Item	Model	INR	Rationale
1	Calibrated scale	Bench scale, 30 kg, 1 g resolution, certified	12,000	Mass is a hard limit; 1 g resolves 0.02 % of the aircraft, so build-up is tracked per component. Certified, so the figure is defensible.
1	Thrust stand	20 kg load cell, HX711 amplifier, power meter, frame in-house	8,000	The motors ship without a thrust curve. Measuring thrust and current together gives thrust per watt. Commercial stands start near INR 45,000.
2	Motor (one)	Tarot TL96020, 5008, 340 KV	4,500	One measured before eleven are committed. Requirement 3.18 kgf, hover at half throttle where efficiency peaks.

Per aircraft, avionics and sensing — phases 3–6, repeated at 13–16 and 21–24 INR 85,200 of parts per aircraft

Ph.	Item	Model	INR	Rationale
3, 13, 21	Flight controller	Holybro Pixhawk 6C Mini	22,600	Two independent motion sensors, so one sensor failure does not end the flight. Runs ArduPilot, which our simulation already targets. Cheaper boards at INR 6,000 lack the memory for our logging and failsafe configuration.
5, 15, 23	GNSS receiver	RTK-capable, RTCM3, $\leq$3 cm CEP	18,000	The largest single source of position error: 3.9 m on ordinary satellite positioning against 0.75 m with RTK. Survivor coordinates are the system's output, so this is not a line to economise on.
6, 16, 24	Camera and lens	Arducam IMX477, 6 mm CS, fixed focus	11,100	At 40 m each pixel covers 1.03 cm of ground, putting 39 pixels across a person in water — enough to detect. Focus is fixed because everything beyond 4.15 m is already sharp, and a focus motor is one more part that can drift unnoticed.
4, 14, 22	AI accelerator	Pi AI HAT+ 26 TOPS (Hailo-8)	11,000	Runs the detector at 130–160 frames per second against the 96 the search needs. The half-speed variant saves INR 4,000 but manages only 60–80, below requirement.
6, 16, 24	Radio and storage	ExpressLRS 2.4 GHz, firmware 3.5+	8,500	Carries pilot control and aircraft data over one link, which removes a second radio costing INR 19,000. Needs firmware 3.5 or later; earlier versions cannot share the link.
4, 14, 22	Onboard computer	Raspberry Pi 5, 8 GB	8,000	Runs the mission planner, the coverage logic and the delivery logic. The only board in its class with the PCIe connector the accelerator plugs into.

Ph.	Item	Model	INR	Rationale
6, 16, 24	Video and coordination radios	Analog 5.8 GHz video set, SX1262 865 MHz modules	6,000	**Open item.** The mesh this replaced had 8.7 dB of margin at 600 m — the weakest link in the system carrying the most data — so it was dropped for three separate radios. This amount is the old mesh node's price carried forward, not a quote for the replacement. Needs pricing before phase 6.

Per aircraft, airframe and drive — phases 7–10, repeated at 17–20 and 25–28 INR 74,100 of parts per aircraft

Ph.	Item	Model	INR	Rationale
10, 20, 28	Motors	Tarot TL96020, 340 KV, 4 off	18,000	3.18 kgf each, giving twice the aircraft weight in thrust. 340 KV suits an 18 in propeller at this pack voltage.
7, 17, 25	Airframe	25 × 23 mm carbon tube, machined clamps, in-house	13,000	The carbon tube is loaded to 4 % of what it can take, so bending is not the risk. The joints are, so clamps are machined rather than printed.
8, 18, 26	Battery cells	Molicel P45B 21700, 4500 mAh, 18 off	12,600	292 Wh, covering the mission, a full second search and four minutes of holding. Each cell peaks at 38.3 A, so a 40 A cell runs at 96 % of rating — which is why a lower-rated cell will not do.
8, 18, 26	Pack assembly	6S 60 A BMS, distribution board, regulator	8,500	Cells arrive loose; covers welding, fusing, balancing and monitoring.
9, 19, 27	Speed controllers	50–60 A continuous, 6S, 4 off	7,200	Each motor peaks at 29 A, so these run at about half their rating. Undersizing here is a known cause of in-flight failure.
9, 19, 27	Propellers	Tarot 1855 carbon, CW/CCW, 4 off	6,500	18 in carbon at INR 3,250 per pair. The most frequently replaced item in a flight campaign.
9, 19, 27	Payload release	Four metal-gear servos, mechanical detents	4,500	Detents rather than holding torque, so a power interruption cannot release a kit.
9, 19, 27	Wiring	10 AWG silicone, XT90-S, JST-GH	2,400	Cable thickness is set by the 115 A peak. The anti-spark connector stops the contacts pitting when a 292 Wh pack is plugged in.
9, 19, 27	Propeller adapters	4 mm bore self-tightening hubs, 4 off	1,400	Rotation seats the nut rather than loosening it; a released propeller loses the aircraft.

Flight-test support — phases 11–12 INR 36,500 of parts

Ph.	Item	Model	INR	Rationale
11	Battery chargers	ToolkitRC M6D, 500 W, 25 A, dual, 2 off	11,672	A 13.5 Ah pack needs about 340 W to charge in an hour. The usual HOTA D6 Pro (INR 10,499–INR 13,500) gives 325 W per channel and would run at its limit; the M6D gives 500 W at INR 5,836 and needs only a supply the department holds. Four channels charge the fleet at once.
12	Consumables	Fasteners, thread-lock, tape, filament	13,000	Consumed continuously in assembly and repair.

Ph.	Item	Model	INR	Rationale
11	Safety-pilot radio	RadioMaster Boxer, ExpressLRS	6,955	Manual override, required every flight. Full-size gimbals: taking control of a 6.4 kg aircraft under fault conditions demands precise input.
12	Sun hood and monitor	Fabricated hood, second-hand monitor	1,500	Laptop displays are illegible in direct sun.

Ground segment and statutory — phases 29–32 INR 52,000 of parts

Ph.	Item	Model	INR	Rationale
31	Field consumables	Cables, mounts, spares, transport protection	25,000	Corrections travel over the existing telemetry link, so no separate radio is needed.
29	Base station	Second GNSS receiver, airborne class	18,000	A second receiver, kept still on a surveyed point. It is what makes RTK work; without it survivor positions degrade to 3.9 m.
32	Registration	Statutory, three aircraft	5,000	At 6.36 kg these are Small category under the Drone Rules 2021.
30	Survey tripod	Photographic tripod and adapter	4,000	A stable, repeatable mount; the corrections are relative, so survey-grade accuracy is unnecessary.

What this arrangement costs us

Thirty-two approvals is a material administrative overhead, and buying one aircraft at a time places the same imported order three times — roughly eight to twelve weeks of schedule. Consolidating the per-aircraft phases into one order is available if schedule becomes binding.

The detection evaluation needs no procurement: about a week on a departmental GPU and 30 GB of storage, against a public dataset.

7. Future work

Near term. Measure what is currently calculated: detection accuracy, motor thrust, setup time. Evaluate near-infrared sensing — water absorbs strongly and skin does not, so a person appears bright against dark water. A camera without its infrared filter costs INR 3,000, and a public dataset allows evaluation before purchase. Only the training data is flood-specific; the same system suits earthquake, wildfire and avalanche search.

Longer term. A capable aircraft deciding for itself within a small power budget is a hard requirement in planetary aviation. *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density; *Dragonfly* launches in 2028 for Titan. A command takes minutes to arrive, so every decision is made on board. What transfers is what is not flood-specific: onboard survey planning, perception run to an energy budget, and safe behaviour without a link. Microcontroller-class processors, often proposed for such missions, cannot resolve targets of this size, so a planetary configuration needs a different operating point.

8. Conclusion

Three aircraft, one operator, no network: survivors located in flooded terrain and reached with a relief kit. Eight minutes for ten hectares, returning coordinates a rescue coordinator can act on. The design is specified; what remains is to build it and measure it.

Exposure is never more than one phase. The first is INR 31,371 and settles whether these motors lift this aircraft; if they do not, the programme stops there and the department keeps a thrust stand and a certified scale. Every later release is backed by the previous result, and the two largest come only after a complete aircraft has flown.

Floods recur every monsoon, and no capability of this kind exists at a scale two people can deploy. The whole system costs less than an hour of helicopter search time. We ask the department to release Phase 1.